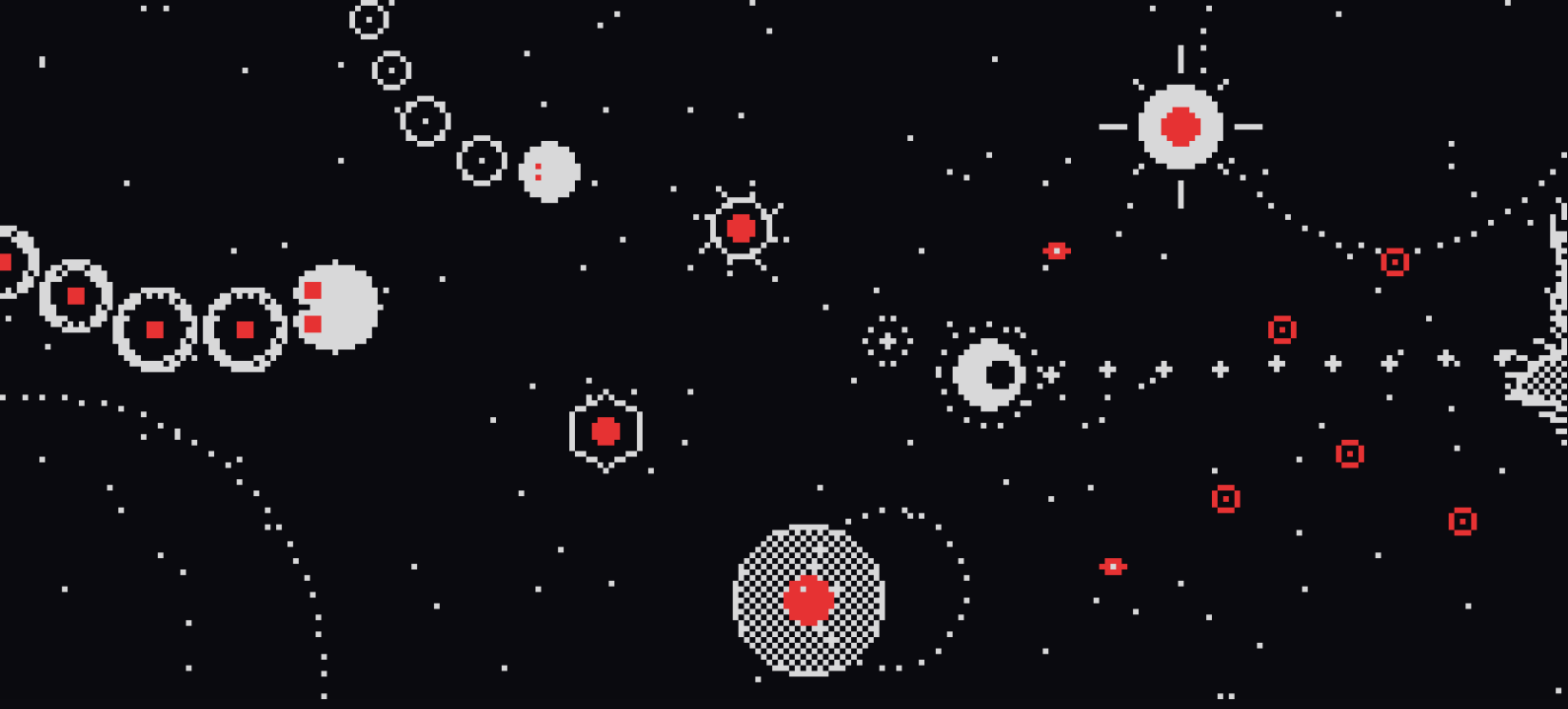


ANTIBACTERIA

Game Design Document

Twin-stick shooter · PlayStation 5, Xbox Series X|S, Nintendo Switch, PC

Abelix · Giorgi Abelashvili

Version 1.0 · November 2026

Contents

01 Game Overview

02 Story and Setting

03 Core Gameplay

04 Game Modes

05 Weapons, Bonuses and Progression

06 Enemies and Bosses

07 Roguelike Perks

08 Art Direction

09 User Interface and Accessibility

10 Technical Overview

11 Business and Production

> **Document purpose.** This document describes AntiBacteria as currently built (version 1.0, November 2026) for platform submission: what the game is, how it plays, its content, its technology and the planned platform features. Items marked `planned` are not yet implemented.

01 Game Overview

AntiBacteria is a fast-paced pixel-art twin-stick shooter set under a microscope. The player is a single living cell that has woken up in a doctor's forgotten sample, and has to blast its way through swarms of bacteria, dodge bullet-hell patterns and take down giant bosses. Short, intense levels; a 50-level campaign; a 50-room roguelike mode with perks and a companion; and three arcade modes built for score chasing.

Key facts

Title	AntiBacteria
Genre	Twin-stick shooter / arena shooter with roguelike mode (bullet hell elements)
Players	1 (single player, offline)
Platforms	PlayStation 5, Xbox Series X|S, Nintendo Switch, PC (Steam). Same-day release planned on all platforms.
Target release	Spring 2027
Business model	Premium, one-time purchase. No microtransactions, no ads, no paid DLC planned.
Engine	Unity 6 (6000.3), Universal Render Pipeline 2D
Languages	15: English, French, German, Italian, Spanish (Spain), Portuguese (Portugal), Dutch, Polish, Czech, Russian, Japanese, Korean, Simplified Chinese, Traditional Chinese, Georgian (text only)
Expected age rating	Low (all ages / 7+ expected). Abstract cartoon violence against microbes; no blood, no gore, no language, no online interaction. Final rating via the platform rating process.
Session length	1–2 minutes per campaign level; 20–60 minutes per roguelike run
Developer	Abelix (Giorgi Abelashvili)

Pitch

"Under the microscope, he left a sample behind. And in the dark, something woke up."

AntiBacteria takes the immediate, readable action of classic arcade twin-stick shooters and wraps it in a bold, strict three-colour pixel-art style: white for everything you can touch, red for everything that matters (danger and reward). Every enemy has a unique silhouette and behaviour, so the screen stays readable even at its busiest.

Unique selling points

- **Readable chaos.** A strict 3-colour visual language (background, white, red) keeps dense swarms and bullet patterns easy to read; four switchable palettes include a colour-blind safe one.
- **Five modes, one tight core.** Campaign, Roguelike, Capture the Flag, Rush and Peace Walker all reuse the same move-aim-shoot core in different ways.
- **Deep roguelike runs.** 50 rooms, 34 perks across four rarities, synergies and a Germ Buddy companion with its own upgrade tree.
- **27 distinct microbes.** Chasers, dodgers, dashers, snakes that split, flowers that bloom into homing petals, turrets and five bosses.
- **Slow motion on demand.** A rechargeable time-slow ability turns near misses into precise dodges.
- **A game with history.** A remake of the developer's own 2013 Mac game, rebuilt from scratch with the original balance preserved.

02 Story and Setting

The story is told in a short, skippable pixel-art intro (about 30 seconds, hold any button to skip) the first time the game is launched. It can be replayed from the menu.

Beat	On screen
1	Late at night, one lab still had its lights on.
2	For years, the doctor had searched for a cure.
3	At last, he called it a day and went home.
4	Under the microscope, he left a sample behind.
5	And in the dark, something woke up.
Title	**You are the cure.** Play begins inside the sample.



Intro frames, beats 1–5.

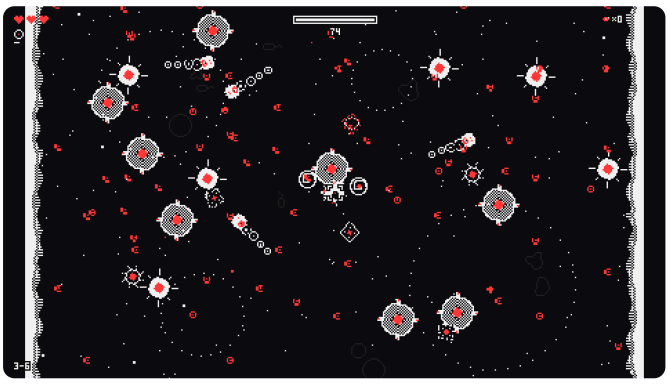
Setting

The whole game takes place inside the sample on the microscope slide. The campaign travels through five regions, each with its own enemy mix and a boss at the end:

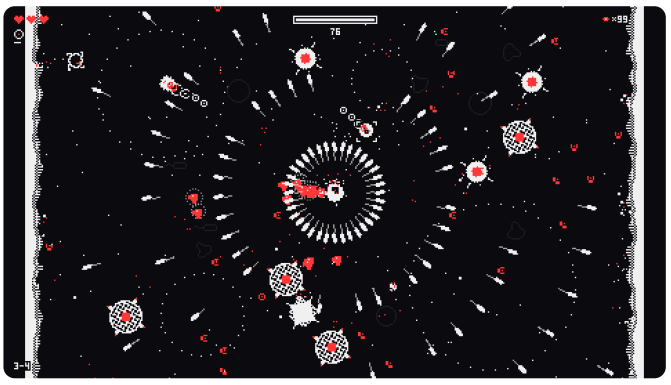
World	Levels	Boss (level)
1. Agar Plains	1–10	BigBoss (10)
2. Fever Vents	11–20	Inferno (20)
3. Crimson Current	21–30	BigRedNoise (30)
4. The Coils	31–40	Anakonda (40)
5. The Nucleus	41–50	Frunctus (50, final boss)

Tone: tense but playful. There is no gore: enemies pop into dotted rings and a few pixels of debris. The player never harms anything that isn't a microbe.

In play



Bullet patterns from spinning turrets.



Circle Shot bonus clearing a crowd.

03 Core Gameplay

Core loop

Move, aim, shoot, survive. The player cell moves freely around a walled arena while aiming independently. Enemies spawn in waves at the arena edges; killing them drops **vitamins** (currency) and, in Roguelike, **XP orbs**. Power-up **bonuses** appear regularly. Levels end when the clock runs out (normal levels) or the boss is defeated.

The player cell

- **Lives:** 3 hearts, restored at the start of each campaign level. A hit costs one heart and gives 2 seconds of invulnerability.
- **Movement:** analogue, 8-way equivalent on keys; constant top speed, no inertia, so dodging is precise.
- **Shooting:** continuous fire while the button is held (optional Auto-Shoot). The volley widens as the campaign goes on: 1, 2, 3 and 4 bullets from levels 1, 11, 21 and 31.
- **Slow motion:** one button slows the whole game to 35% speed for 10 seconds, then recharges for 5 seconds. Not available on boss levels.

Controls

Action	DualSense (PS5)	Keyboard and mouse (PC)
Move	Left stick (or d-pad)	W A S D / arrow keys
Aim	Right stick	Mouse
Shoot	R2 (or R1)	Left mouse button
Slow motion	L2 (or L1)	Right mouse button / Space
Pause	Options button	Esc / P
Menus	Cross to confirm, Circle to go back	Enter / Esc, mouse

All controls can be remapped in Options, separately for keyboard and gamepad. On-screen button prompts follow the device in use (PlayStation, Xbox, Switch or keyboard glyphs).

First-play tutorial

On the first level ever played, after "Get Ready", a step-by-step prompt teaches **Move**, **Aim and Shoot** and **Slow Motion** with the player's real bindings. Each step waits until the player has done it; enemies only start coming after the last step. It shows once and is saved.

Level structure (campaign)

- **Normal levels:** survive an 80-second clock. Waves arrive on a timer from the arena edges; the number of enemies per wave grows during the level.
- **Boss levels** (every 10th): one boss, no clock, a health bar at the top of the screen.
- A level is won by surviving or killing the boss; any leftover vitamins fly to the player. Losing all hearts ends the level (Retry or return to the map).

04 Game Modes

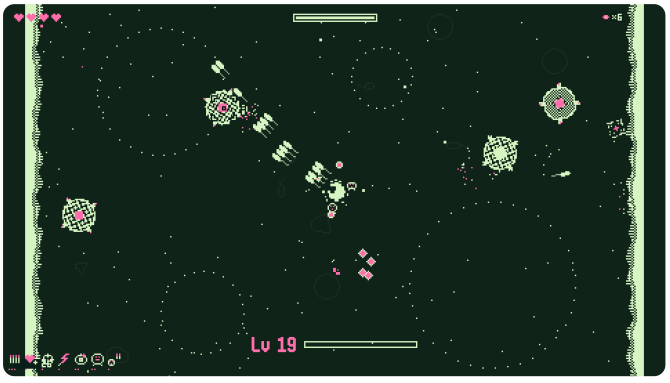
Campaign

50 levels across 5 worlds, unlocked in order on a world map. Each level is a hand-tuned mix of enemy types, wave timing and wave size (balance carried over unchanged from the 2013 original). Every level can be replayed on three difficulties, with a best score per level and difficulty.

Roguelike

A run of up to **50 rooms** in a row, built on the campaign's levels and bosses. Each room sends a set number of enemies; when they are all dead, a gate opens in the top wall to the next room. Lives carry over between rooms.

- **XP and perks:** kills drop XP orbs. Filling the XP bar pauses the game and offers **3 perk cards**; the player picks one. Perks stack in ranks.
- **Rarity:** Common, Rare, Epic and Legendary. Deeper rooms (and harder difficulties) roll rarer perks.
- **Germ Buddy:** a perk that adds a small companion bacterium that follows the player and shoots; it unlocks seven buddy-only upgrades.
- **Scaling:** rooms grow in size, enemy health grows per room, boss rooms every 10 rooms restore one life.
- **Fair runs:** shop upgrades don't apply in Roguelike; every run starts the same and perks are the only power.



Roguelike: a build with extra bullet lines and the Germ Buddy (Petri Dish palette).



Level up: choose one of three perks.

Arcade modes (endless, for high scores)

Mode	Goal	Twist
Capture the Flag	Reach flags that appear around the arena while waves keep coming.	Every few flags the waves get faster and bigger and the volley grows.
Rush	Stand inside zones until they fill up.	You have to hold your ground in one spot while enemies close in.
Peace Walker	Collect flags before a 16.7-second timer runs out.	No gun at all. Dodge indestructible tentacles; one more spawns with every flag.

Difficulty

Difficulty	Changes	Rewards
Normal	Classic pace	Score ×1, vitamins ×1
Hard	Faster waves, quicker enemies, tougher bosses	Score ×2, vitamins ×1.5
Impossible	Relentless swarms, the fastest bosses	Score ×3, vitamins ×2

05 Weapons, Bonuses and Progression

Shot types

Shot	Pattern	How to get it
Normal	Fast single stream, widening to 4 bullets as the campaign goes on	Default
Water	Very fast stream	Bonus pickup, or buy in the shop
Windmill	Eight bullets in two rotating crosses	Bonus pickup, or buy in the shop
Circle	A full ring of 36 bullets	Bonus pickup, or buy in the shop
Tornado	A spinning three-way spiral	Bonus pickup, or buy in the shop

Bonuses

In normal campaign levels a bonus appears every 20 seconds, away from the player, and wanders for about 11 seconds before it disappears. A boxed icon cycles through the bonus types until picked up. Timed bonuses last 13.3 seconds.

Bonus	Effect
Water / Windmill / Circle / Tornado Shot	Temporary weapon
+1 Life	Restores a heart (up to 3)
Booooooom!	A blast that deals 20 hits to every enemy nearby
Immortality	No damage for the duration
Double Speed	Twice the movement speed

Vitamins and the upgrade shop

Every kill drops vitamins (red capsules) that fly to the player when close; harder difficulties pay more. Vitamins are kept between sessions and spent in **Upgrade Character** on permanent upgrades:

Upgrade	Price (vitamins)
Water, Windmill, Circle or Tornado as the base shot (each)	20,000
Double Speed (toggle)	40,000
Immortality (toggle)	1,000,000

Upgrades apply to the campaign and the arcade modes, not to Roguelike. Only vitamins earned in play can be spent; there are no paid currencies.

Rewards and meta progression

- Best score per level and difficulty; campaign total; best room per difficulty in Roguelike.
- Five online leaderboards: Campaign, Capture the Flag, Rush, Peace Walker, Roguelike.
- 20 achievements / trophies (section 10).

06 Enemies and Bosses

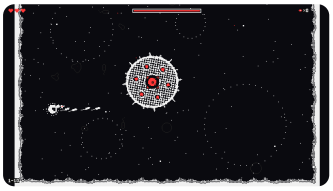
27 enemy types in total (including bosses, minions and snake segments). Every type has its own silhouette and movement so it can be recognised in a single colour. New types are introduced gradually across the five worlds.

Enemy	Behaviour	Hits
BlueCorosia	Basic drifting cell; hurts on contact only.	1
Loctopus	Small chaser that flies straight at the player.	1
RedNoise	Red chaser that homes from the moment it appears; also the minion of BigRedNoise.	1
SpiralCorosia	Advances while tracing big loops, hard to hit.	1
ZigZagus	Homing cell that wobbles side to side.	1
Adrenalium	Hyperactive dasher that rockets between points, often where the player stood.	1
BigBlueCorosia	Large, slow, spinning wanderer.	7
BigHunterus	Big, slow, tanky blob that always heads for the player.	5
ShootingCorosia	Big spinning wanderer that fires 4-way bursts.	10
Sunux	Spinning 'sun' that stays at the arena edge and sprays 8-way bullet bursts.	20
Chorux	Red core protected by two orbiting defenders that block bullets until the core is hit.	1
Gnida	Homer that dodges sideways when you aim at it.	1
ZamriGnida	Looping Gnida variant that dodges even harder.	1
Serpentus	Fast snake that charges; its body blocks shots, aim for the head.	7 / segment
ShootingSerpentus	Shorter snake whose head fires aimed bullets.	4 / segment
CaterPillar	Long worm; shoot the head and the whole worm pops.	10
LotusBacterius	A snake that blooms: killing its head releases six fast homing petals.	10 head, 1 petal
Spawner	Wandering nest that keeps releasing small enemies (arcade modes).	10
Tentacle	Indestructible wanderer, the hazard of Peace Walker.	–

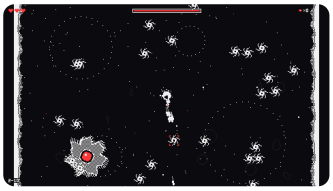
Bosses

Boss	Level	Behaviour
BigBoss	10	A screen-filling spinning blob that roams fast and charges where you stand. No bullets: its size and speed are the threat.
Inferno	20	A huge spinning mass that speeds up as it takes damage and fills the arena with invulnerable tentacles to dodge.
BigRedNoise	30	A giant red blob that chases you; at three health thresholds it becomes invulnerable and births streams of red chasers.
Anakonda	40	A giant, very fast snake of 24 segments that shoots aimed bullets and keeps growing as new segments race in to join its tail.
Frunctus	50	The final boss: a dodging, dashing blob with three phases of bullet fans, lunges, spirals and a brood of dodging minions.

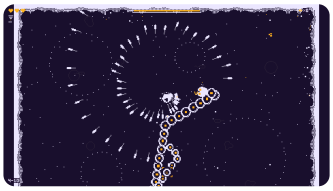
Boss health and attack timings scale with difficulty (for example, Inferno sends 10, 20 or 25 tentacles on Normal, Hard and Impossible).



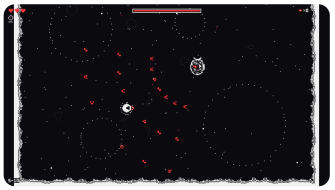
BigBoss



Inferno



Anakonda



Frunctus

07 Roguelike Perks

34 perks. Each level up offers three different cards; each card first rolls a rarity (weighted by how deep the run is), then a perk of that rarity that can still rank up. The Reroll perk lets the player redraw the cards.

Player perks

Perk	Rarity	Max rank	Effect
Rapid Fire	Common	3	Shoot 12% faster per rank
Extra Bullet	Rare	3	+1 parallel bullet line, up to 4
Heavy Shots	Epic	2	+1 damage per bullet
Piercing	Rare	2	Bullets go through 1 more enemy
Quick Feet	Common	4	Move 12% faster
Vitality	Common	2	+1 max life, heals 1
First Aid	Rare	1	Heal 1 life every 3 rooms
Shield	Rare	1	Blocks 1 hit in every room
Second Wind	Epic	1	Survive one deadly hit
Nova	Epic	1	Losing a life blasts nearby enemies
Magnet	Common	3	Pull orbs from 50% farther
Wisdom	Common	3	+15% XP from orbs
Golden Touch	Common	3	+50% vitamins
Bonus Lover	Common	3	Bonuses last 40% longer
Time Bender	Common	2	Slow motion recharges 50% faster
Critical Hit	Common	3	10% chance of double damage
Big Shots	Common	1	Bullets are 2× bigger
Reroll	Common	3	+1 card reroll at every level up
Ricochet	Rare	2	Bullets bounce off walls
Frost Shots	Rare	2	15% chance to slow enemies you hit
Lucky	Rare	2	Rarer perks show up more often
Chain Zap	Epic	2	Kills zap 1 nearby enemy
Orbiter	Epic	2	A cell circles you, hurting enemies
Vampire	Epic	1	Heal 1 life every 60 kills
Last Stand	Legendary	1	Shoot 2× faster on your last life
Overdrive	Legendary	1	Every 10th volley: a huge piercing blast
Germ Buddy	Rare	2	A small bacterium follows you and shoots

Germ Buddy upgrades

Offered only once the Germ Buddy perk is owned.

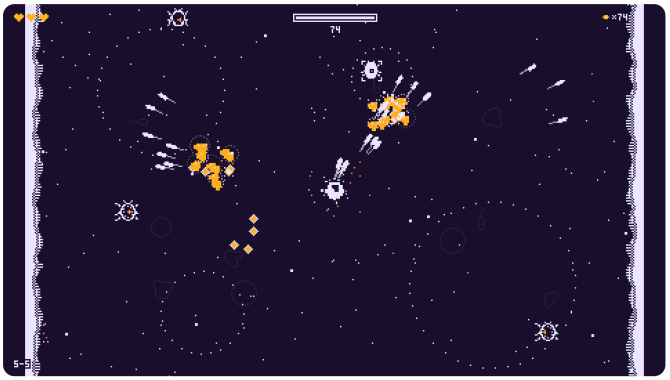
Perk	Rarity	Max rank	Effect
Buddy Trigger	Common	2	Buddy shoots 15% faster
Buddy Punch	Common	2	Buddy shots deal +25% damage
Fetch	Common	1	Buddy pulls orbs and vitamins to you
Double Barrel	Rare	1	Buddy fires +1 bullet line
Sharpshooter	Rare	2	Buddy shots pierce 1 more enemy
Bodyguard	Epic	1	Once a room, the buddy takes a hit for you
Copycat	Epic	1	Buddy copies your Frost, Ricochet and Critical

08 Art Direction

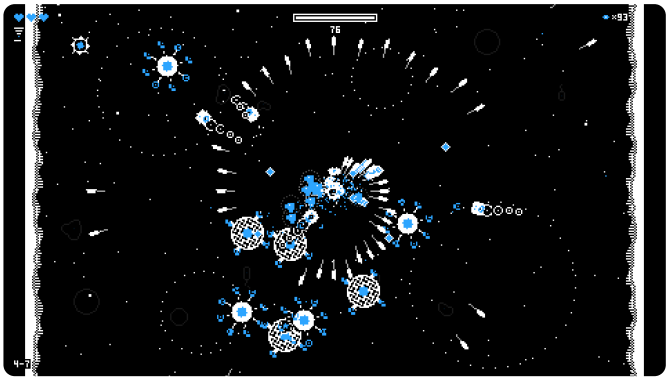
Downwell-inspired pixel art with a strict three-colour palette, where colour means role and shape tells things apart.

Palette index	Role	Used for
0	Background	The microscope field
1	Foreground (white)	The player, the player's shots, enemy bodies, the arena wall, neutral effects
2	Accent (red)	Anything that needs attention: enemy cores, enemy bullets, vitamins, hearts

- **Red means danger or reward; white is everything else.** Red things are told apart by shape: ring = enemy bullet, capsule = vitamin, heart = life.
- **Every enemy has a unique silhouette** that reads in one colour, which is what makes the colour-blind palette work.
- **Pixel-perfect:** the game renders a low-resolution pixel grid and scales it by whole numbers only, so pixels stay crisp at every resolution (4× at 1080p, 9× at 4K; the view widens slightly on screens that aren't an exact multiple).
- **Palettes:** Classic, Petri Dish, Gram Stain and Colour-blind, switchable at any time from Options.
- **Effects:** dotted ring shockwaves, sparks and screen shake instead of gore.



Gram Stain palette.



Colour-blind palette.

Audio

Synthesized chiptune music and sound effects in keeping with the pixel art: a level theme, a boss theme, menu music and an intro score. Sound effects are layered so rapid fire, kills and pickups stay readable (for example, collecting vitamins in a row climbs in pitch). Separate music and sound volume settings.

09 User Interface and Accessibility

Flow

Intro (first launch) → Main menu (Start Game, Upgrade Character, Leaderboards, Options, About, Quit) → mode select → world map and level select (campaign) or difficulty → gameplay → result screen (Level Clear / Game Over with score, kills, vitamins, time) → next level, retry or menu. A pause menu is available at any time and the game pauses itself when the system menu or an overlay opens.

HUD

- Hearts (top left), level clock and slow-motion meter (top centre), vitamin counter (top right), level number (bottom left).
- Boss levels show a boss health bar; Roguelike shows the XP bar, level and owned perks.
- Off-screen bonuses are pointed out with an arrow at the screen edge.

Accessibility

Feature	Details
Colour-blind palette	One of four palettes; every gameplay object is also distinguished by shape.
Full remapping	Keyboard and gamepad bindings, with a reset to defaults.
Auto-Shoot	Optional: fire continuously so only moving and aiming are needed.
Difficulty choice	Normal, Hard and Impossible per level/run.
Readable text	Large pixel font sized for TV distance; all menus usable with a controller only.
Skippable story	The intro can be skipped and replayed later.
Tutorial	Step-by-step first-play controls tutorial using the player's own bindings.
Localization	15 languages, switchable in Options; dedicated pixel fonts for Japanese, Korean, Chinese and Georgian.

10 Technical Overview

Engine	Unity 6 (6000.3), URP 2D, Pixel Perfect Camera
Input	Rewired (console-ready input layer, remapping, per-family button glyphs)
Localization	Gley Localization + TextMeshPro, 15 languages
Rendering target (PS5)	Up to 3840×2160 output with whole-number pixel scaling; 60 fps target
Performance	2D sprites only; very low GPU and memory use. Object pooling for bullets and effects.
Save data	One small save file (campaign progress, scores, wallet, upgrades, settings). On PS5: saved through the platform save data system, autosave after each level and on quit.
Online	No online multiplayer. Online leaderboards and trophies only; the game is fully playable offline.
Platform layer	All store features go through one platform interface; each platform plugs in its own backend (Steam on PC, PSN on PS5).
Install size	Small (well under 1 GB expected)

PlayStation 5 features

Feature	Plan
Trophies	Trophy set based on the 20 achievements below (final list and grades to follow Sony's trophy guidelines). planned
DualSense adaptive triggers	R2 resistance while firing (varies by shot type); L2 click for slow motion. planned
DualSense haptics	Distinct haptic feedback for shooting, hits, kills, bonuses and boss attacks. planned
Activities	Activity cards for each world and its boss, the Roguelike run and each arcade mode. planned
Leaderboards	Five online leaderboards (Campaign, Capture the Flag, Rush, Peace Walker, Roguelike). planned
Fast loading	Near-instant level loads (small 2D scenes).
Pause on system menu	Gameplay pauses when the PS button / system overlay is used (already done for the Steam overlay). planned

Achievements / trophies

Group	Unlocked by
World bosses (4)	Beating the boss of level 10, 20, 30 and 40
Final boss	Beating level 50
Hard / Impossible	Clearing any level on Hard / on Impossible
First Upgrade / All Upgrades	Buying an upgrade / owning every upgrade
Vitamin Hoarder	Holding 1,000 vitamins
Capture the Flag / Rush / Peace Walker	25 flags / 10 zones / 10 flags in one run
Roguelike rooms (3)	Clearing room 10, 25 and all 50 rooms
Roguelike Impossible	Clearing room 20 on Impossible
Germ Buddy / Legendary / Four Lines	Taking the Germ Buddy, a Legendary perk, reaching 4 bullet lines

11 Business and Production

Monetization

Premium one-time purchase. No microtransactions, loot boxes, ads or paid currencies. The original 2013 version's real-money "Vitamin Store" has been removed; vitamins are earned only by playing. Price to be announced.

Target audience

- Fans of arcade twin-stick shooters and bullet-hell games looking for short, intense sessions.
- Roguelite players who enjoy build-crafting and "one more run".
- Score chasers (leaderboards, arcade modes, three difficulties).
- Pixel-art and retro fans; families and younger players (no gore, simple controls, Normal difficulty and tutorial).

Status and schedule

Milestone	Status / target
Core game (all 5 modes, 50 levels, 5 bosses, 50-room roguelike)	Complete, in polish and balancing
Localization (15 languages)	Complete, native review pending
PC (Steam) build, achievements, leaderboards, cloud saves	Complete
Steam demo (Next Fest)	February 2027 (target)
PS5, Xbox Series X|S and Switch ports and certification	Winter–spring 2027
Release	Spring 2027, same day on all platforms (target)

Team

Abelix: Giorgi Abelashvili (game design, programming, art direction). AntiBacteria is a remake of the developer's own 2013 Mac game of the same name, rebuilt from scratch in Unity for modern platforms.

Third-party technology

Unity 6; Rewired (Guavaman Enterprises); Gley Localization; m6×11 font by Daniel Linssen (with the Cyrillic extension by Natto Games); Noto Sans CJK (SIL Open Font License) and DejaVu Sans, re-rasterized as pixel fonts.